

Hugo Lyons Keenan

hugo0076@gmail.com

Personal Statement

I am a PhD candidate researching AI safety and trustworthiness at the University of Melbourne. My work focuses on detecting and understanding anomalous behaviours in neural networks, motivated by a broader desire to ensure AI systems are reliable and beneficial. I have experience communicating technical research to diverse audiences and thrive in collaborative, interdisciplinary environments. Over my career, I aim to contribute to both technical AI safety research and to the communication of its importance to broad audiences.

Education History

The University of Melbourne, Doctor of Philosophy Jul 2025 - Present

- Researching mechanistic approaches to detecting anomalous behaviour in neural networks, with applications to identifying backdoors and other safety-relevant failure modes in frontier AI systems.
- Motivated by a broader goal of developing the basic science that will enable safe deployment of increasingly capable AI systems.

The University of Melbourne, Master of Computer Science, *First Class Honours* Feb 2022 – Oct 2024

- Thesis on adversarially robust out-of-distribution detection resulted in a paper accepted at SaTML 2025, a competitive international venue for ML research.
- Developed a novel, theoretically principled detection method with applications in AI safety; demonstrated improvements over six baselines across multiple threat models.
- Thesis awarded a mark of 89/100.

The University of Melbourne, Bachelor of Science, *First Class Honours* Feb 2019 – Oct 2021

- Completed coursework in probability, statistics, computer science, economics, and philosophy.
- Achieved second-highest mark in final subject (Applied Data Science).

Experience

Research Assistant, NLP Group, The University of Melbourne May 2024 – Jul 2024

- Constructed and curated a large-scale news corpus; performed data quality validation and preliminary linguistic analysis.
- Dataset published under open-source licence and used for ongoing research.

Research Assistant, Agent-Based Modelling Group, The University of Melbourne Nov 2023 – Apr 2024

- Calibrated agent-based models to real-world reversal learning data from behavioural experiments.
- Developed a parallelised codebase for efficient experimentation across large parameter spaces.
- Project resulted in a paper accepted at the Journal of Artificial Societies and Social Simulation (JASSS).

Academic Tutor / Head Tutor, The University of Melbourne Feb 2022 - Present

- Tutored/Head Tutored 2–4 classes of ~25 students per semester in first year, third year, and master's level computer science.
- Received overwhelmingly positive feedback, with all students responding either 'Agree' or 'Strongly Agree' to questions about tutorial usefulness, quality of teaching and tutor availability in an anonymous survey.

Quantitative Trading Intern, Akuna Capital Dec 2021 - Feb 2022

- Used machine learning techniques to discover trends in exchange data. Developed and back tested numerous trading strategies based on these trends. Wrote scripts to adjust internal volatility metrics to market information in real time.
- Received an offer of full-time employment.

Awards/Scholarships

Research Training Program Scholarship, UniMelb

AUD \$ 38500 / year

The Research Training Program Scholarship is provided by the Australian Commonwealth Government and the University of Melbourne and is awarded to high-achieving graduate researchers who are enrolled in a graduate research degree.

CIS PhD Top-up Scholarship, UniMelb

AUD \$ 20000 / year

The Computing and Information Systems PhD Scholarship is offered to high achieving graduates of the University of Melbourne Master of Computer Science who are enrolling in the Doctor of Philosophy – Engineering and Information Technology in Computing and Information Systems.

Extracurricular Activities

President/Treasurer, One for the World UniMelb

Aug 2020 – Sep 2022

- Managed a university club aimed at empowering students to make highly effective donations to global health charities.
- Relevant skills: Community outreach, management of finances, setting meetings, communication with team members, delegation of tasks.

Marketing Officer, Effective Altruism UniMelb

Feb 2021 – Aug 2022

- Worked with other members of the group to produce marketing materials for activities, panels and recruitment.
- Relevant skills: Graphic design, communication with team members, deadline management.

Publications

- **Hugo Lyons Keenan**, Christopher Leckie, Sarah Erfani. "Mechanistic Anomaly Detection via Functional Attribution." International Conference on Machine Learning (ICML) 2026
- **Hugo Lyons Keenan**, Sarah Erfani, Christopher Leckie. "HALO: Robust Out-of-Distribution Detection via Joint Optimisation." 2025 IEEE Conference on Secure and Trustworthy Machine Learning (SaTML). IEEE, 2025.
- Roben Delos Reyes, **Hugo Lyons Keenan**, Cameron Zachreson. "A Microsimulation Model of Behaviour Change Calibrated to Reversal Learning Data." Journal of Artificial Societies and Social Simulation (JASSS) 28 (3) 8